27)

$f(z) = z^2 - iz + 3 - 2i$ $\quad z_0 = 2 - i$

$$\lim_{z \to z_0} f(z) = \lim_{z \to 2-i} (z^2 - iz + 3 - 2i)$$

$$f(z_0) = (2-i)^2 - i(2-i) + 3 - 2i$$

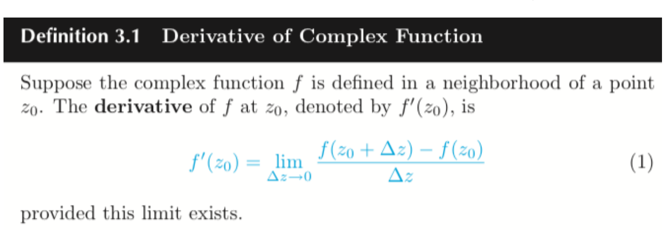
$$f(2-i) = 4 - 4i + i^2 - 2i + i^2 + 3 - 2i$$
$$= 7 - 8i - 1 - 1$$

$$f(2-i) = 5 - 8i$$

Hence $f(2-i)$ exist, So, the function $f(z) = z^2 - iz + 3 - 2i$ is continous at the point $z_0 = 2 - i$

3.1) Example 1)

$f(z) = z^2 - 5z$



Definition 3.1 Derivative of Complex Function

Suppose the complex function f is defined in a neighborhood of a point z_0. The **derivative** of f at z_0, denoted by $f'(z_0)$, is

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \tag{1}$$

provided this limit exists.

firstly;

$$f(z + \Delta z) = (z + \Delta z)^2 - 5(z + \Delta z) = z^2 + 2z\Delta z + (\Delta z)^2 - 5z - 5\Delta z$$

Secondly;

$$f(z + \Delta z) - f(z) = z^2 + 2z\Delta z + (\Delta z)^2 - 5z - 5\Delta z - (z^2 - 5z)$$
$$= 2z\Delta z + (\Delta z)^2 - 5\Delta z$$

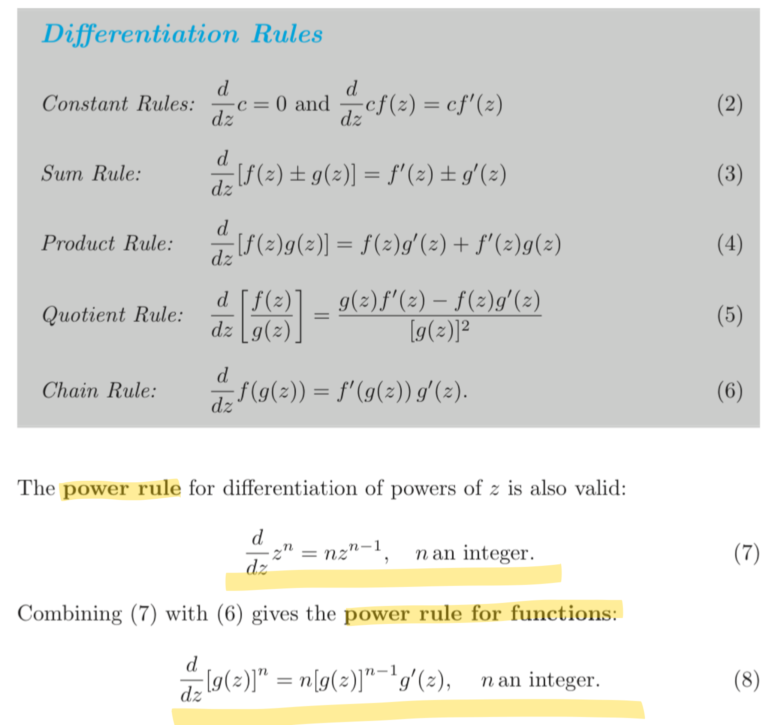
finally;

$$f'(z) = \lim_{\Delta z \to 0} \frac{2z\Delta z + (\Delta z)^2 - 5\Delta z}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{\Delta z (2z + \Delta z - 5)}{\Delta z}$$

$$= \lim_{\Delta z \to 0} (2z + \Delta z - 5)$$

The limit $f'(z) = 2z - 5$



Differentiation Rules

Constant Rules: $\frac{d}{dz}c = 0$ and $\frac{d}{dz}cf(z) = cf'(z)$ (2)

Sum Rule: $\frac{d}{dz}[f(z) \pm g(z)] = f'(z) \pm g'(z)$ (3)

Product Rule: $\frac{d}{dz}[f(z)g(z)] = f(z)g'(z) + f'(z)g(z)$ (4)

Quotient Rule: $\frac{d}{dz}\left[\frac{f(z)}{g(z)}\right] = \frac{g(z)f'(z) - f(z)g'(z)}{[g(z)]^2}$ (5)

Chain Rule: $\frac{d}{dz}f(g(z)) = f'(g(z))\,g'(z).$ (6)

The **power rule** for differentiation of powers of z is also valid:

$$\frac{d}{dz}z^n = nz^{n-1}, \quad n \text{ an integer.} \tag{7}$$

Combining (7) with (6) gives the **power rule for functions**:

$$\frac{d}{dz}[g(z)]^n = n[g(z)]^{n-1}g'(z), \quad n \text{ an integer.} \tag{8}$$

Ex2)

a) $f(z) = 3z^4 - 5z^3 + 2z$

$$f'(z) = 3.4z^3 - 5.3z^2 + 2.1 = 12z^3 - 15z^2 + 2$$

b) $f(z) = \frac{z^2}{4z+1}$

$$f'(z) = \frac{(4z+1).2z - z^2.4}{(4z+1)^2} = \frac{4z^2 + 2z}{(4z+1)^2}$$

c) $f(z) = (iz^2 + 3z)^5$

$n = 5$ $\quad g(z) = iz^2 + 3z$ $\quad g'(z) = 2iz + 3$

$$f'(z) = 5(iz^2 + 3z)^4(2iz + 3)$$

Ex 3)

Show that function $f(z) = x + 4iy$ is not differentiable at any point z

$$\Delta z = \Delta x + i\Delta y$$

$$f(z + \Delta z) - f(z) = (x + \Delta x) + 4i(y + \Delta y) - x - 4iy = \Delta x + 4i\Delta y$$

$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{\Delta x + 4i\Delta y}{\Delta x + i\Delta y}$$

$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{\Delta x}{\Delta x} = 1$$

$$\lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \to 0} \frac{4i\Delta y}{i\Delta y} = 4$$

f isn't differentiable at any point z.

Ex 4)

Compute $\lim_{z \to 2+i} \frac{z^2 - 4z + 5}{z^3 - z - 10i}$

$f(z) = z^2 - 4z + 5$ $\quad g(z) = z^3 - z - 10i$

for $z = 2+i$ $\quad f(z)$ and $g(z)$ equal to zero

so;

$$\lim_{z \to 2+i} \frac{z^2 - 4z + 5}{z^3 - z - 10i} = \frac{f'(2+i)}{g'(2+i)} = \frac{2z - 4}{3z^2 - 1} = \frac{2i}{8 + 12i} = \frac{3}{26} + \frac{1}{13}i$$

---

1) 1) $f(z) = 9iz + 2 - 3i$

$f'(z) = 9i$

9) $f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$

$$f(z) = z^4 - z^2$$

$$\lim_{z \to z_0} \frac{z^4 - z^2 - (z_0^4 - z_0^2)}{z - z_0}$$

$$= z^4 - z^2 - z_0^4 + z_0^2$$

$$= \frac{(z^2 - z_0^2)(z^2 + z_0^2) - (z - z_0)(z + z_0)}{z - z_0}$$

$$= \frac{\cancel{(z - z_0)}(z + z_0)(z^2 + z_0^2) - \cancel{(z - z_0)}(z + z_0)}{\cancel{z - z_0}}$$

$$\lim_{z \to z_0} (z + z_0)(z^2 + z_0^2 - 1)$$

$$(2z_0)(2z_0^2 - 1)$$

---

14) $f(z) = (z^2 + 2z - 7i)^2(z^4 - 4iz)^3$

$$f'(z) = 2.(z^2 + 2z - 7i).(2z + 2)(z^4 - 4iz^3) + 3.(z^4 - 4iz)^2(4z^3 - 4i)[z^2 + 2z - 7i]^2$$

---

18) $f(z) = \left(\frac{(4+2i)z}{(2-i)z^2 + 9i}\right)^3$

$$f'(z) = 3.\left(\frac{(4+2i)z}{(2-i)z^2 + 9i}\right)^2 . \left(\frac{[(4+2i).z]' . [(2-i)z^2 + 9i] - [(2-i)z^2 + 9i]' . [(4+2i)z]}{[(2-i)z^2 + 9i]^2}\right)$$

$$f'(z) = 3.\left(\frac{(4+2i)z}{(2-i)z^2 + 9i}\right)^2 . \frac{(4+2i).((2-i)z^2 + 9i) - [(4-2i)z . ((4+2i)z)]}{[(2-i)z^2 + 9i]^2}$$

---

23) $\lim_{z \to i} \frac{z^7 + i}{z^{14} + 1} = \frac{0}{0}$ $\quad$ use L'hosp. Rule

$$\frac{7z^6}{14z^{13}} = \frac{1}{2 . z^7}$$

24) $\lim_{z \to \sqrt{2} + \sqrt{2}i} \frac{z^4 + 16}{z^2 - 2\sqrt{2}z + 4} = \frac{0}{0}$

$$\frac{4z^3}{2z - 2\sqrt{2}} = \frac{2z^3}{z - \sqrt{2}}$$

---

27) $f(z) = \frac{iz^2 - 2z}{3z + 1 - i}$ $\to \neq 0$

$3z + 1 - i \neq 0$

$z = \frac{i-1}{3}$ only this point not analytic

28) $f(z) = (z^4 - 2iz^2 + z)^{10}$

all the point is analytic